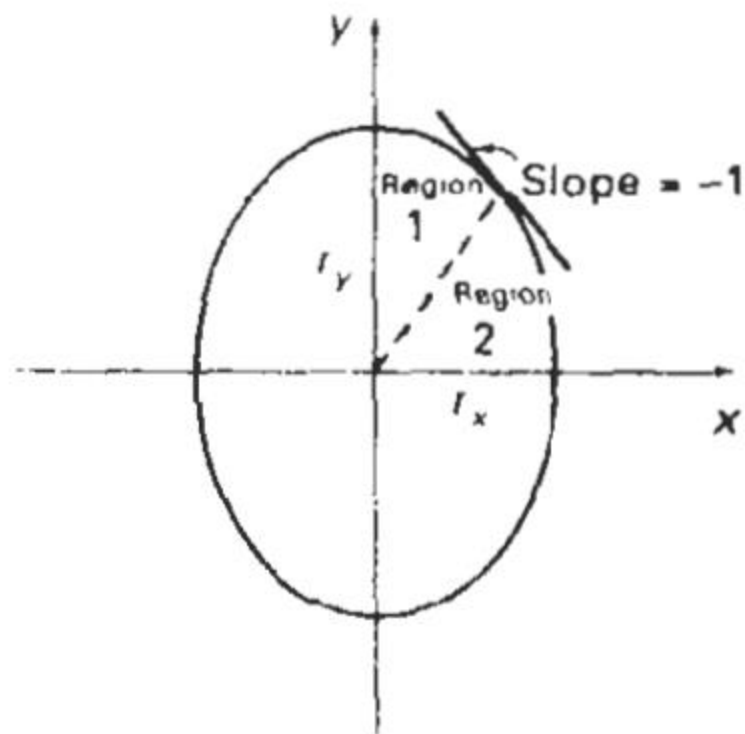


## MID-POINT-ELLIPSE ALGORITHM

$$\left(\frac{x - x_c}{r_x}\right)^2 + \left(\frac{y - y_c}{r_y}\right)^2 = 1$$



$$f_{\text{ellipse}}(x, y) = r_y^2 x^2 + r_x^2 y^2 - r_x^2 r_y^2$$

which has the following properties:

$$f_{\text{ellipse}}(x, y) \begin{cases} < 0, & \text{if } (x, y) \text{ is inside the ellipse boundary} \\ = 0, & \text{if } (x, y) \text{ is on the ellipse boundary} \\ > 0, & \text{if } (x, y) \text{ is outside the ellipse boundary} \end{cases}$$

SLOPE OF ELLIPSE IS

$$\frac{dy}{dx} = -\frac{2r_y^2 x}{2r_x^2 y}$$

At the boundary between region 1 and region 2,  $dy/dx = -1$  and

$$2r_y^2 x = 2r_x^2 y$$

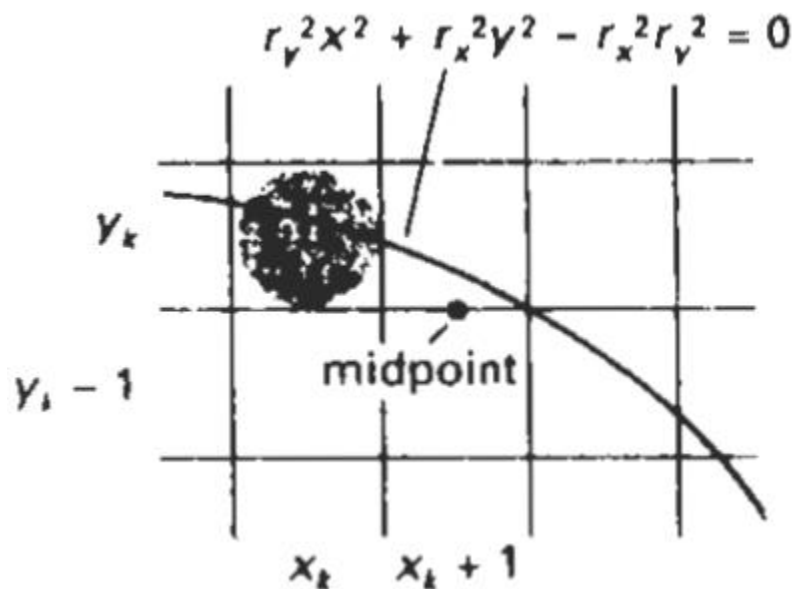
Therefore, we move out of region 1 whenever

$$2r_y^2 x \geq 2r_x^2 y$$

$$p1_k = f_{\text{ellipse}}\left(x_k + 1, y_k - \frac{1}{2}\right)$$

$$= r_y^2(x_k + 1)^2 + r_x^2\left(y_k - \frac{1}{2}\right)^2 - r_x^2 r_y^2$$

If  $p1_k < 0$ , the midpoint is inside the ellipse and the pixel on scan line  $y_k$  is closer to the ellipse boundary. Otherwise, the midposition is outside or on the ellipse boundary, and we select the pixel on scan line  $y_k - 1$ .



$$\begin{aligned}
 p1_{k+1} &= f_{\text{ellipse}}\left(x_{k+1} + 1, y_{k+1} - \frac{1}{2}\right) \\
 &= r_y^2[(x_k + 1) + 1]^2 + r_x^2\left(y_{k+1} - \frac{1}{2}\right)^2 - r_x^2 r_y^2
 \end{aligned}$$

or

$$p1_{k+1} = p1_k + 2r_y^2(x_k + 1) + r_y^2 + r_x^2\left[\left(y_{k+1} - \frac{1}{2}\right)^2 - \left(y_k - \frac{1}{2}\right)^2\right]$$

Decision parameters are incremented by the following amounts:

$$\text{increment} = \begin{cases} 2r_y^2 x_{k+1} + r_y^2, & \text{if } p1_k < 0 \\ 2r_y^2 x_{k+1} + r_y^2 - 2r_x^2 y_{k+1}, & \text{if } p1_k \geq 0 \end{cases}$$

At the initial position  $(0, r_y)$ ,

$$2r_y^2x = 0$$

$$2r_x^2y = 2r_x^2r_y$$

In region 1, the initial value of the decision parameter is obtained by evaluating the ellipse function at the start position  $(x_0, y_0) = (0, r_y)$ :

$$\begin{aligned} p1_0 &= f_{\text{ellipse}}\left(1, r_y - \frac{1}{2}\right) \\ &= r_y^2 + r_x^2\left(r_y - \frac{1}{2}\right)^2 - r_x^2r_y^2 \end{aligned}$$

or

$$p1_0 = r_y^2 - r_x^2r_y + \frac{1}{4}r_x^2$$

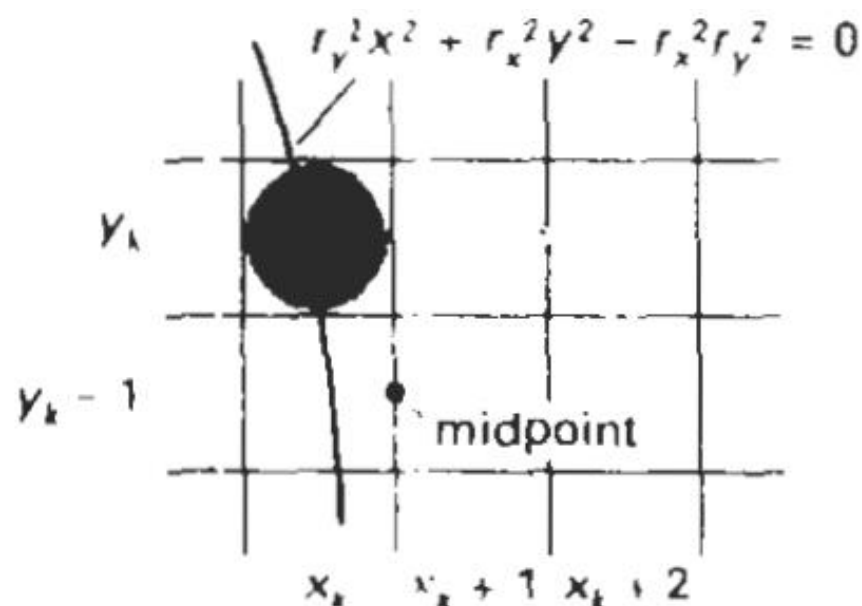
Over region 2, we sample at unit steps in the negative  $y$  direction, and the midpoint is now taken between horizontal pixels at each step

For this region, the decision parameter is evaluated as

$$p2_k = f_{\text{ellipse}}\left(x_k + \frac{1}{2}, y_k - 1\right)$$

$$= r_y^2\left(x_k + \frac{1}{2}\right)^2 + r_x^2(y_k - 1)^2 - r_x^2 r_y^2$$

If  $p2_k > 0$ , the midposition is outside the ellipse boundary, and we select the pixel at  $x_k$ . If  $p2_k \leq 0$ , the midpoint is inside or on the ellipse boundary, and we select pixel position  $x_{k+1}$ .



$$\begin{aligned}
 p2_{k+1} &= f_{\text{ellipse}}\left(x_{k+1} + \frac{1}{2}, y_{k+1} - 1\right) \\
 &= r_y^2\left(x_{k+1} + \frac{1}{2}\right)^2 + r_x^2[(y_k - 1) - 1]^2 - r_x^2 r_y^2
 \end{aligned}$$

or

$$p2_{k+1} = p2_k - 2r_x^2(y_k - 1) + r_x^2 + r_y^2\left[\left(x_{k+1} + \frac{1}{2}\right)^2 - \left(x_k + \frac{1}{2}\right)^2\right]$$

When we enter region 2, the initial position  $(x_0, y_0)$  is taken as the last position selected in region 1 and the initial decision parameter in region 2 is then

$$\begin{aligned}
 p2_0 &= f_{\text{ellipse}}\left(x_0 + \frac{1}{2}, y_0 - 1\right) \\
 &= r_y^2\left(x_0 + \frac{1}{2}\right)^2 + r_x^2(y_0 - 1)^2 - r_x^2 r_y^2
 \end{aligned}$$

To simplify the calculation of  $p2_0$ , we could select pixel positions in counterclockwise order starting at  $(r_x, 0)$ . Unit steps would then be taken in the positive  $y$  direction up to the last position selected in region 1.

## Midpoint Ellipse Algorithm

1. Input  $r_x$ ,  $r_y$ , and ellipse center  $(x_c, y_c)$ , and obtain the first point on an ellipse centered on the origin as

$$(x_0, y_0) = (0, r_y)$$

2. Calculate the initial value of the decision parameter in region 1 as

$$p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$$

3. At each  $x_k$  position in region 1, starting at  $k = 0$ , perform the following test: If  $p1_k < 0$ , the next point along the ellipse centered on  $(0, 0)$  is  $(x_{k+1}, y_k)$  and

$$p1_{k+1} = p1_k + 2r_y^2 x_{k+1} + r_y^2$$

Otherwise, the next point along the circle is  $(x_k + 1, y_k - 1)$  and

$$p1_{k+1} = p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2$$

with

$$2r_y^2 x_{k+1} = 2r_y^2 x_k + 2r_y^2, \quad 2r_x^2 y_{k+1} = 2r_x^2 y_k - 2r_x^2$$

and continue until  $2r_y^2 x \geq 2r_x^2 y$ .

4. Calculate the initial value of the decision parameter in region 2 using the last point  $(x_0, y_0)$  calculated in region 1 as

$$p2_0 = r_y^2 \left( x_0 + \frac{1}{2} \right)^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$$

- At each  $y_k$  position in region 2, starting at  $k = 0$ , perform the following test: If  $p2_k > 0$ , the next point along the ellipse centered on  $(0, 0)$  is  $(x_k, y_k - 1)$  and

$$p2_{k+1} = p2_k - 2r_x^2 y_{k+1} + r_x^2$$

Otherwise, the next point along the circle is  $(x_k + 1, y_k - 1)$  and

$$p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2$$

using the same incremental calculations for  $x$  and  $y$  as in region 1.

- Determine symmetry points in the other three quadrants.
- Move each calculated pixel position  $(x, y)$  onto the elliptical path centered on  $(x_c, y_c)$  and plot the coordinate values:

$$x = x + x_c \quad y = y + y_c$$

- Repeat the steps for region 1 until  $2r_y^2 x \geq 2r_x^2 y$ .

Given input ellipse parameters  $r_x = 8$  and  $r_y = 6$ , we illustrate the steps in the midpoint ellipse algorithm by determining raster positions along the ellipse path in the first quadrant. Initial values and increments for the decision parameter calculations are

$$2r_y^2x = 0 \quad (\text{with increment } 2r_y^2 = 72)$$

$$2r_x^2y = 2r_x^2r_y \quad (\text{with increment } -2r_x^2 = -128)$$

For region 1: The initial point for the ellipse centered on the origin is  $(x_0, y_0) = (0, 6)$ , and the initial decision parameter value is

$$p1_0 = r_y^2 - r_x^2r_y + \frac{1}{4}r_x^2 = -332$$

Successive decision parameter values and positions along the ellipse path are calculated using the midpoint method as

| $k$ | $p1_k$ | $(x_{k+1}, y_{k+1})$ | $2r_y^2x_{k+1}$ | $2r_x^2y_{k+1}$ |
|-----|--------|----------------------|-----------------|-----------------|
| 0   | -332   | (1, 6)               | 72              | 768             |
| 1   | -224   | (2, 6)               | 144             | 768             |
| 2   | -44    | (3, 6)               | 216             | 768             |
| 3   | 208    | (4, 5)               | 288             | 640             |
| 4   | -108   | (5, 5)               | 360             | 640             |
| 5   | 288    | (6, 4)               | 432             | 512             |
| 6   | 244    | (7, 3)               | 504             | 384             |

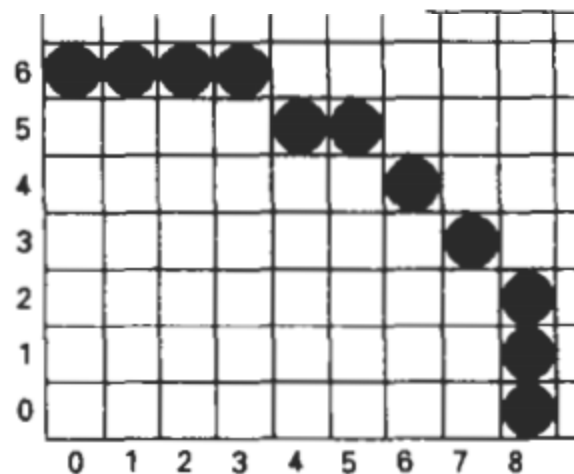
We now move out of region 1, since  $2r_y^2x > 2r_x^2y$ .

For region 2, the initial point is  $(x_0, y_0) = (7, 3)$  and the initial decision parameter is

$$p2_0 = f\left(7 + \frac{1}{2}, 2\right) = -151$$

The remaining positions along the ellipse path in the first quadrant are then calculated as

| $k$ | $p2_k$ | $(x_{k+1}, y_{k+1})$ | $2r_y^2x_{k+1}$ | $2r_x^2y_{k+1}$ |
|-----|--------|----------------------|-----------------|-----------------|
| 0   | -151   | (8, 2)               | 576             | 256             |
| 1   | 233    | (8, 1)               | 576             | 128             |
| 2   | 745    | (8, 0)               | —               | —               |



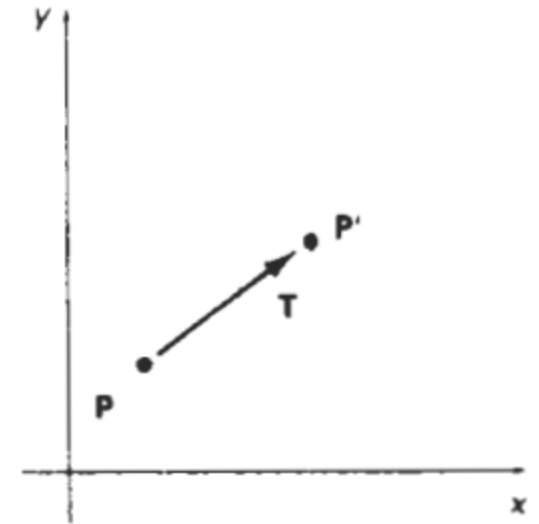
## 2 dimensional transformations

Translation

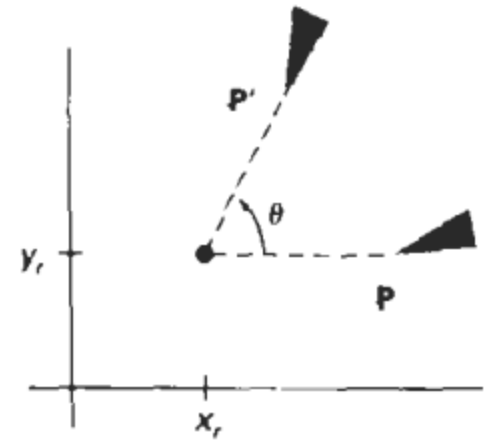
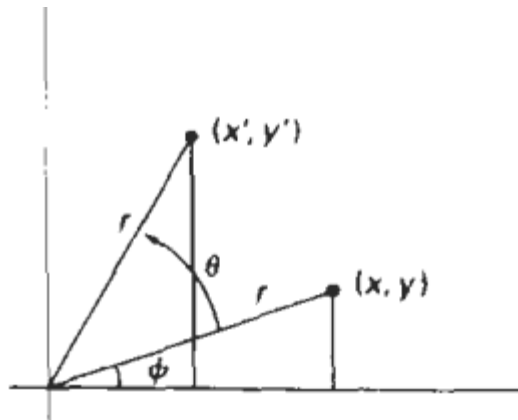
$$x' = x + t_x, \quad y' = y + t_y$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}$$

$$\mathbf{P} = [x \ y] \text{ and } \mathbf{T} = [t_x \ t_y]$$



## Rotation



$$x' = r \cos (\phi + \theta) = r \cos \phi \cos \theta - r \sin \phi \sin \theta$$

$$y' = r \sin (\phi + \theta) = r \cos \phi \sin \theta + r \sin \phi \cos \theta$$

$$x = r \cos \phi, \quad y = r \sin \phi$$

$$x' = x \cos \theta - y \sin \theta$$

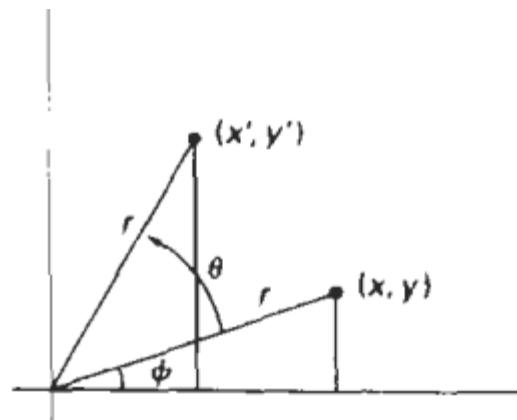
$$y' = x \sin \theta + y \cos \theta$$

$$\mathbf{P}' = \mathbf{R} \cdot \mathbf{P}$$

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

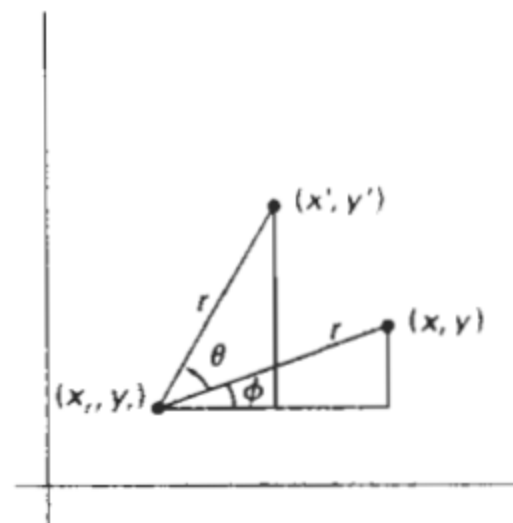
$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$



$$x' = x_r + (x - x_r) \cos \theta - (y - y_r) \sin \theta$$

$$y' = y_r + (x - x_r) \sin \theta + (y - y_r) \cos \theta$$



## Scaling



(a)

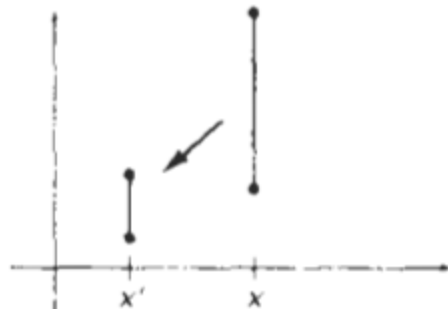


(b)

$$x' = x \cdot s_x, \quad y' = y \cdot s_y$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{S} \cdot \mathbf{P}$$



$$x' = x_f + (x - x_f)s_x, \quad y' = y_f + (y - y_f)s_y$$

$$x' = x \cdot s_x + x_f(1 - s_x)$$

$$y' = y \cdot s_y + y_f(1 - s_y)$$

$$\mathbf{P}' = \mathbf{T}(t_x, t_y) \cdot \mathbf{P}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}$$

$$\mathbf{P} = [x \ y] \text{ and } \mathbf{T} = [t_x \ t_y]$$

$$\mathbf{P}' = \mathbf{S}(s_x, s_y) \cdot \mathbf{P}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{S} \cdot \mathbf{P}$$

$$\mathbf{P}' = \mathbf{R}(\theta) \cdot \mathbf{P}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{P}' = \mathbf{R} \cdot \mathbf{P}$$

$$\mathbf{R} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

## COMPOSITE TRANSFORMATIONS

### Translations

$$\begin{aligned}\mathbf{P}' &= T(t_{x2}, t_{y2}) \cdot \{T(t_{x1}, t_{y1}) \cdot \mathbf{P}\} \\ &= \{T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1})\} \cdot \mathbf{P}\end{aligned}$$

$$\begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y1} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

or

$$T(t_{x2}, t_{y2}) \cdot T(t_{x1}, t_{y1}) = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$$

which demonstrates that two successive translations are additive.

Rotations

$$\begin{aligned}\mathbf{P}' &= \mathbf{R}(\theta_2) \cdot \{\mathbf{R}(\theta_1) \cdot \mathbf{P}\} \\ &= \{\mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1)\} \cdot \mathbf{P}\end{aligned}$$

By multiplying the two rotation matrices, we can verify that two successive rotations are additive:

$$\mathbf{R}(\theta_2) \cdot \mathbf{R}(\theta_1) = \mathbf{R}(\theta_1 + \theta_2)$$

so that the final rotated coordinates can be calculated with the composite rotation matrix as

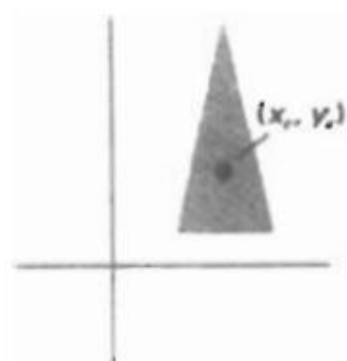
$$\mathbf{P}' = \mathbf{R}(\theta_1 + \theta_2) \cdot \mathbf{P}$$

## Scalings

$$\begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_{x1} \cdot s_{x2} & 0 & 0 \\ 0 & s_{y1} \cdot s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

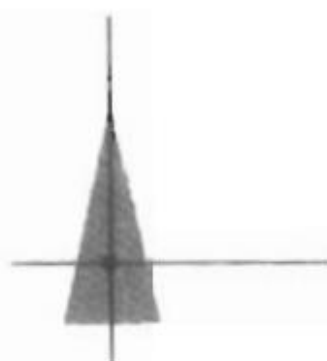
$$\mathbf{S}(s_{x2}, s_{y2}) \cdot \mathbf{S}(s_{x1}, s_{y1}) = \mathbf{S}(s_{x1} \cdot s_{x2}, s_{y1} \cdot s_{y2})$$

The resulting matrix in this case indicates that successive scaling operations are multiplicative. That is, if we were to triple the size of an object twice in succession, the final size would be nine times that of the original.



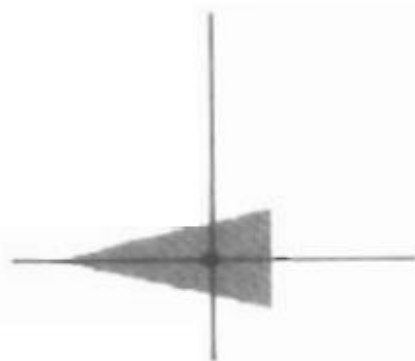
(a)

Original Position  
of Object and  
Pivot Point



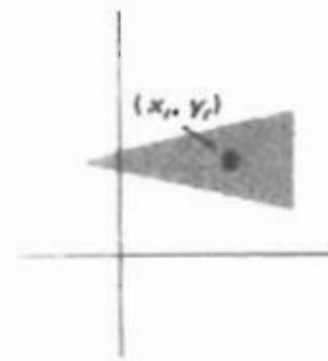
(b)

Translation of  
Object so that  
Pivot Point  
( $x_r, y_r$ ) is at  
Origin



(c)

Rotation  
about  
Origin



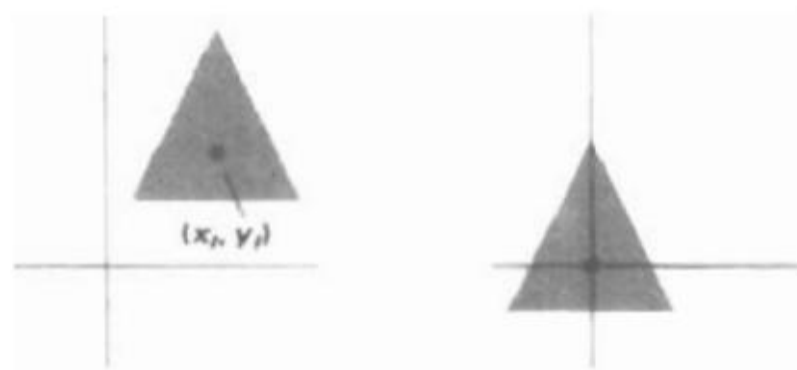
(d)

Translation of  
Object so that  
the Pivot Point  
is Returned  
to Position  
( $x_r, y_r$ )

$$\begin{bmatrix} 1 & 0 & x_r \\ 0 & 1 & y_r \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_r \\ 0 & 1 & -y_r \\ 0 & 0 & 1 \end{bmatrix} \\
 = \begin{bmatrix} \cos \theta & -\sin \theta & x_r(1 - \cos \theta) + y_r \sin \theta \\ \sin \theta & \cos \theta & y_r(1 - \cos \theta) - x_r \sin \theta \\ 0 & 0 & 1 \end{bmatrix}$$

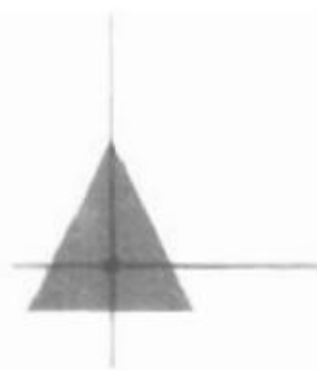
which can be expressed in the form

$$T(x_r, y_r) \cdot R(\theta) \cdot T(-x_r, -y_r) = R(x_r, y_r, \theta)$$



(a)

Original Position  
of Object and  
Fixed Point



(b)

Translate Object  
so that Fixed Point  
( $x_f, y_f$ ) is at Origin



(c)

Scale Object  
with Respect  
to Origin



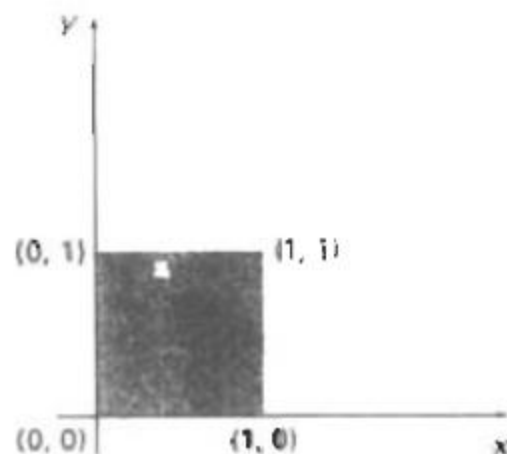
(d)

Translate Object  
so that the Fixed Point  
is Returned to  
Position ( $x_f, y_f$ )

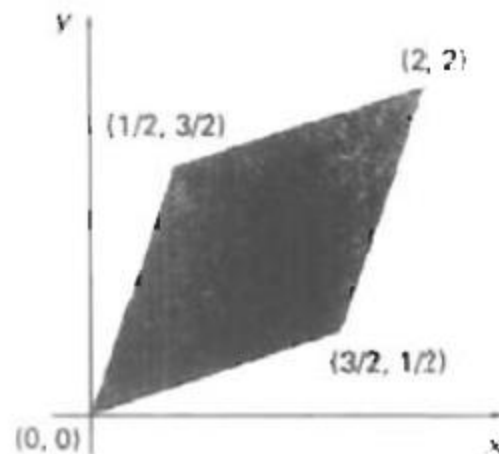
$$\begin{bmatrix} 1 & 0 & x_f \\ 0 & 1 & y_f \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -x_f \\ 0 & 1 & -y_f \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & x_f(1 - s_x) \\ 0 & s_y & y_f(1 - s_y) \\ 0 & 0 & 1 \end{bmatrix}$$

OR

$$T(x_f, y_f) \cdot S(s_x, s_y) \cdot T(-x_f, -y_f) = S(x_f, y_f, s_x, s_y)$$



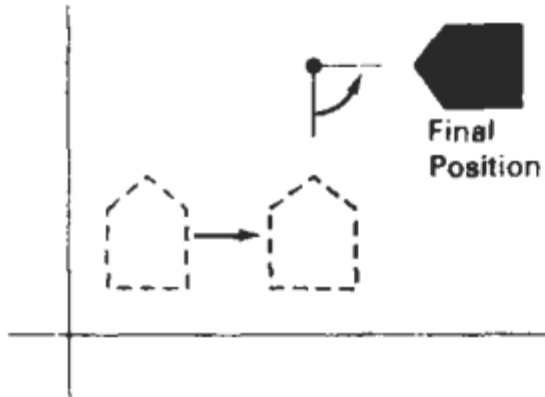
(a)



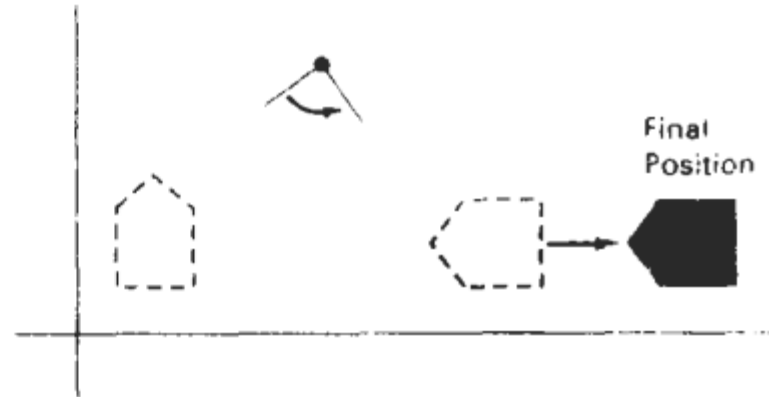
(b)

$$\mathbf{R}^{-1}(\theta) \cdot \mathbf{S}(s_1, s_2) \cdot \mathbf{R}(\theta) = \begin{bmatrix} s_1 \cos^2 \theta + s_2 \sin^2 \theta & (s_2 - s_1) \cos \theta \sin \theta & 0 \\ (s_2 - s_1) \cos \theta \sin \theta & s_1 \sin^2 \theta + s_2 \cos^2 \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We rotate the diagonal onto the  $y$  axis and double its length with the transformation parameters  $\theta = 45^\circ$ ,  $s_1 = 1$ , and  $s_2 = 2$ .



(a)



(b)

Reversing the order in which a sequence of transformations is performed may affect the transformed position of an object. In (a), an object is first translated, then rotated. In (b), the object is rotated first, then translated.

